Chapter 1 – Defining and Collecting Data

Section 1.1 – Defining Data

Categorical (qualitative) variables – have values that can only be placed into categories, such as "yes" and "no". These are typically word responses.

Ex. – asking about: gender, eye color, major
and even # answers that can't be computed
like age & zip code
in asking the question

Numerical (quantitative) variables – have values that represent quantities. These are number responses.

Ex. – a # answer; age, weight, gpa

- Discrete variables – numerical values that arise from a counting process. These are typically whole or integer values.
  sub type

  Ex. – # of siblings, # of books
  anything that is counted

- Continuous variables – numerical values that arise from a measuring process. They can be decimal values, even if they are not expressed that way.
  sub type

  Ex. - weight, height, speed, time, volume
  anything that is measured

Section 1.2 – Measurement Scales for Variables

Measurement Scales:

1.) Nominal scale – classifies data into distinct categories in which no ranking is implied. This is the weakest form of measurement.

normally words

Ex: gender, hair color

2.) Ordinal scale – classifies data into distinct categories in which ranking is implied, but the difference between the categories is not. difference between categories is not quantifyable

Ex: excellent/good/fair/poor
freshman/sophmore/junior/senior – not clear/universal # of credits differentiating between 2 ppl's rank

3.) Interval scale – an ordered scale in which the differences between measurements is a meaningful quantity, but does not have a true zero point.

Ex: temperature

4.) Ratio Scale – an ordered scale in which the difference between the measurements involves a true zero point.

Ex: weight

Section 1.3 – Collecting Data

Primary source of data – when the data collector is the one using the data for analysis.

Ex. – when you collect data and then use the data:
you work for Geico and sample 50% of your clients to then see if you can get them better insurance rates

Secondary source of data – when the person performing the statistical analysis is not the data collector.

Ex. - You just use the data, don't collect:
when we use data from Census Bureau or other internet data

Population – all the items or individuals about which you want to reach conclusions.

Ex. – when it's ALL things in the category:
All Drake Students

Parameter – a measure that describes a characteristics of a population.

Ex. – any # taken from a population is a parameter:
of all Drake students avg gpa is 2.75 (ex not real)

μ (mu): population mean π: population proportion σ (sigma): population standard deviation

Sample – the portion of a population selected for analysis. These are the people or items from which you actually collect data.

Ex. – instead of All items, using a sample of the whole:
37 students in a class vs All Drake students

Statistic – a measure that describes a characteristic of a sample.

any # we get from sample:
of 37 students avg gpa is 3.12

Ex. –

$\bar{x}$: sample mean p : sample proportion s : sample standard deviation

Data Cleaning – may be necessary when you find irregularities in the values collected such as undefined or impossible values. These are different based on categorical or quantitative variables. May include outliers or missing values.

data entry errors: if category is age and entry is 148 – must delete if can't find

outliers: if an entry is much higher/lower & could skew data set (like Oprah in an income set) most likely would delete that entry
or exclude
could transform but not relevant for now

Section 1.4 –Types of Sampling Methods

Frame – a listing of items that make up the population, numbered from 1 to N, from which the sample will be selected.

Ex: list of all Drake students from Registar office

Nonprobability sample – you select the items or individuals without knowing their probabilities of selection.

not characteristic/representative of population

Types of nonprobability sampling:

1.) Convenience sample – items are chosen because they are easy to get a hold of.

Ex. - friends, pple who call into radio show

2.) Judgment sample – you get the opinions of preselected experts in the subject matter.

Ex. - as King finance proffesors what stock to buy

Advantages of nonprobability sampling:

Convienent, cheap, fast, easy

Disadvantages of nonprobability sampling:

lack of accuracy, ppl are biased, results aren't general & only represent the sample

Probability sample – you select the items or individuals based on known probabilities.

Types of probability sampling:

1.) Simple Random sample – every item has the same chance of selection as every other item. Difference between sampling with and without replacement.

Ex. – drawing names from hat

2.) Systematic sample – divide the *N* items into *n* groups of *k* items, *k=N/n*.

a. Start by choosing the first item randomly within the first *k* items in the frame
b. Choose every *k*th item in the frame thereafter

Ex. – I select ever 6th student who walks into class

3.) Stratified sample – divide the items into groups based on a common characteristic and then choose a few items from every group.

Ex. – if we took 50 fr, 50 so, 50 jr, & 50 sr from Drake

4.) Cluster sample – divide the items into groups based on a common characteristic and then choose all the items from a few of the groups.

Ex. – take only so & jrs but All of them

Advantages of probability sampling:

time consuming, costly, difficult to find ppl/track them down

Disadvantages of probability sampling:

Section 1.5 – Types of Survey Errors

We hear about surveys or polls in the media all the time. To identify surveys that lack objectivity or credibility, you need to evaluate the purpose of the survey and by whom it was conducted.

Types of Survey Error:

1.) Coverage Error – when certain groups of items are excluded from the frame so they have no chance of being selected in the sample.

Ex. – telephone survey, not all ppl/groups have a phone

2.) Nonresponse Error – failure to collect data on all items in the sample.

Ex. – mail out 2500 surveys but only 8 ppl send it back

3.) Sampling Error – the variation or "chance differences" that occur simply because different individuals are chosen for different samples.

Unavoidable bc each sample is different, going from class to class to find avg gpa will yield different results due to many factors

Ex. – Any statment like" results of this poll are expected to be within ±3% of actual value"

4.) Measurement Error – includes :

a. ambiguous wording of questions –

like "how much do you study?"

b. the halo effect –

when a person wants to please the interviewer OR when they lie bc they feel uncomfortable revealing the truth

c. respondent error –

when someone is over/under zealus, over zealus could be like halo effect but under zealus could just be ridiculus answers from an annoyed respondent